

More Convergence Tests

MATH 146 – Calculus II

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Integral Test

Theorem (Integral Test)

Suppose that $f(x)$ is a **positive** function and let $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_1^{\infty} f(x) dx \text{ converges}$$

Theorem (p -Series Test)

The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$ and diverges for $p < 1$.

The Comparison Test

Theorem (The Comparison Test)

Suppose $\sum a_n$ and $\sum b_n$ are series such that $a_n \geq 0$ and $b_n \geq 0$ for all n .

- If $\sum b_n$ converges and $0 \leq a_n \leq b_n$, then $\sum a_n$ converges as well.
- If $\sum a_n$ diverges and $0 \leq a_n \leq b_n$, then $\sum b_n$ diverges as well.

Example: The Comparison Test

Example: Does $\sum_{k=1}^{\infty} \frac{1}{k^2 + 14k + 1}$ converge?

Example: Does $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^3}$ converge?

Example: Does $\sum_{k=2}^{\infty} \frac{1}{\ln(k)}$ converge?

Limit Theorems

The usual arithmetic laws apply for infinite series as well.

Theorem

Suppose that $\sum a_n = A$ and $\sum b_n = B$ are convergent series and that c is a constant. Then:

- $\sum (a_n + b_n) = A + B.$
- $\sum (c \cdot a_n) = cA.$

Example: Does $\sum_{k=1}^{\infty} \frac{12 + k}{5k^3 + k + 1}$ converge?

The Ratio Test

Theorem (The Ratio Test)

Suppose $\sum a_n$ is a series such that $a_n > 0$ and let $r = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}$.

- If $r < 1$, then $\sum a_n$ converges.
- If $r > 1$, then $\sum a_n$ diverges.
- If $r = 1$, the test is inclusive – the series could converge or diverge.

Idea: Compare to a geometric series.

Example: The Ratio Test

Example: Does $\sum_{n=1}^{\infty} \frac{1}{n!}$ converge?

Example: Does $\sum_{k=1}^{\infty} k^2 2^{-k}$ converge?

Example: Does $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge?

The Root Test

Theorem (The Root Test)

Suppose $\sum a_n$ is a series such that $a_n \geq 0$ and let $\rho = \lim_{n \rightarrow \infty} \sqrt[n]{a_n}$.

- If $0 \leq \rho < 1$, then $\sum a_n$ converges.
- If $\rho > 1$, then $\sum a_n$ diverges.
- If $\rho = 1$, the test is inclusive – the series could converge or diverge.

Example: Does $\sum_{k=1}^{\infty} \left(\frac{3k-2}{8k+7} \right)^k$ converge?

The Limit Comparison Test

Theorem (The Limit Comparison Test)

Suppose $\sum a_n$ and $\sum b_n$ are series such that $a_n > 0$ and $b_n > 0$ for all n and let $L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

- If $0 < L < \infty$, then $\sum a_n$ and $\sum b_n$ converge or diverge together.
- If $L = 0$ **AND** $\sum b_n$ converges, then $\sum a_n$ converges.
- If $L = \infty$ **AND** $\sum b_n$ diverges, then $\sum a_n$ diverges.

Example: Does $\sum_{k=1}^{\infty} \frac{1}{k^2 - 1}$ converge?

The Alternating Series Test

Definition

An infinite series is said to be **alternating** if it is of the form $\sum (-1)^n a_n$ or $\sum (-1)^{n+1} a_n$, where $a_n \geq 0$.

Theorem (The Alternating Series Test)

Suppose $\sum (-1)^n a_n$ is an alternating series such that a_n is decreasing monotonically to zero. Then $\sum (-1)^n a_n$ converges

Example: Does $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$ converge?

A Divergence Test

Theorem (A Test for Divergence)

If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Theorem (Contrapositive of a Test for Divergence)

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$ diverges.

Example: Does $\sum_{k=1}^{\infty} \frac{k+1}{3k+1}$ converge?